

Pavan Siva Kumar

Somana.

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DATABASE ANALYST · SQL · PYTHON · DATA ANALYTICS · DATA VISUALIZATION · CSE '26'

SUMMARY

Bachelor’s degree in Computer Science Engineering with specialization in SQL, Python, Database Management, and Data Analytics. Experienced in developing software products based on databases with use of PostgreSQL, MySQL, MongoDB, Firebase, and REST APIs. Knowledgeable in cleaning data, Exploratory Data Analysis (EDA), database design, visualization, and automation with use of Python, Pandas, NumPy, and Matplotlib. Experience in development of different software and IoT products with databases, analytics, and Machine Learning as well as publications in IEEE journals.

EDUCATION

2022–2026	B.Tech — Computer Science Engineering Kalasalingam Academy of Research and Education, Tamil Nadu, Krishnan Kovil CGPA 7.76
2020–2022	Class XII — MPC Tirumala Educational Institutions, Bhimavaram 93.7%
2019–2020	10th Standard — SSC Bhashyam School, Bhimavaram 100%

SKILLS

Database & Data Analysis	PostgreSQL, MySQL, MongoDB, Firebase, Supabase & SQL, Pandas, NumPy, Matplotlib, Data Cleaning, EDA, Data Visualization.
Programming	Python, Java, JavaScript, C, C++, Dart
AI/ML	YOLOv8, OpenCV, Computer Vision, CNNs, Object Detection, Image Classification
Tools	Jupyter Notebook, Google Colab, PyTorch, Ultralytics, ONNX, Flutter
IoT	Raspberry Pi, ESP32, Arduino, Sensor Integration, IoT Dashboards.
Development	Git, GitHub, VS Code, Android Studio, Bridge, Blockbench

PROJECTS

- 01

Green Guardian GO — ML-Powered Reverse Vending Machine
Developed an AI-powered reverse vending machine using YOLOv8 and Raspberry Pi for real-time plastic bottle detection and validation. Designed a PostgreSQL database to manage bottle records, user rewards, and machine transactions. Integrated computer vision with IoT hardware to automate recycling workflows.

Raspberry Pi

Python

PostgreSQL

YOLOv8

Computer Vision
- 02

BluNest — Smart Aquarium Care IoT Application
Built an IoT-based aquarium monitoring system for real-time sensor data collection and visualization. Integrated Firebase for live monitoring of temperature, pH, turbidity, TDS, and water-level data. Designed the system architecture for future predictive analytics using historical sensor data.

Flutter App

Firebase

IoT

ESP32

Machine learning
- 03

Dual-Mode Robotic Arm
Built a robotic arm supporting manual joystick control and AI-assisted pick-and-place operations. Secured 1st Place at the IEEE RAS Project Showcase.

Raspberry Pi

Robotics

AI Vision

Motor Control

Sensors
- 04

Facial Recognition-Based Smart Attendance System
Developed a facial recognition attendance system using Python and OpenCV. Automated attendance logging with MongoDB, reducing manual record management.

Python

MongoDB

OpenCV

Face Recognition
- 05

Minecraft Bedrock Addon Development & Game Scripting
Developed custom Minecraft Bedrock addons using JSON-based behavior/resource packs and JavaScript scripting. Designed custom blocks, items, abilities, recipes, textures, and gameplay mechanics while debugging Bedrock API behavior using Bridge and Blockbench.

JavaScript

JSON

Bridge

Blockbench

API Debugging

ACHIEVEMENTS

- 01

Finalist, Top 10 of 500+ teams — Ground Reality, Launchpad '25, E-Cell BITS Pilani Hyderabad
- 02

1st Place — RAS Project Showcase for Dual-Mode Robotic Arm
- 03

Recognized for excellence in competitive engineering events, achieving Top 10 Finalist in two hackathons and securing 6th Place in a Project Expo for innovative software and hardware solutions.
- 04

Completed certifications in DBMS Course(codechef), Oracle Cloud Infrastructure AI, Microsoft Azure, and Cloud IoT, strengthening expertise in SQL, cloud technologies, and data-driven application development.

IEEE PUBLICATIONS

- 01

[BluNest: IoT & ML-Based Intelligent Aquarium Management System](#)
- 02

[Fix My Ride: ML-Enabled Platform for Intelligent Automotive Assistance](#)
- 03

[Facial Recognition-Based Smart Attendance System Using MongoDB and OpenCV](#)
- 04

[Dual-Mode Robotic Arm for Manufacturing Industries](#)
- 05

[IoT Enabled Device-to-Device Communication for Smart City Application](#)